

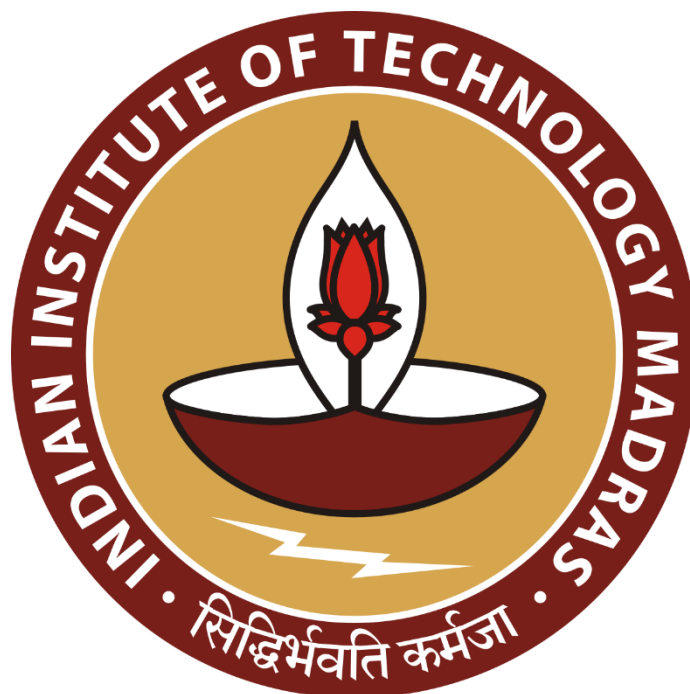
Daily sales analysis of a café to increase efficiency and profits

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “Daily sales analysis of a cafe to increase efficiency and profits”. I extend my appreciation to **Coffee Beans**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: Vedaant Arora

Date: May 13, 2025

1 Executive Summary and Title

The project is focused on studying daily sales data for providing suggestions/solutions to problems faced by Coffee Beans, A café located at Anand Vihar, Delhi. The business works on a B2C model, serving people that walk in to order and also via delivery through a shop hired delivery boy.

The major challenge that ‘Coffee Beans’ faces is that the daily sales fluctuate by a lot, this makes it very difficult to forecast expected sales or revenue. This leads to over purchasing and under purchasing of raw materials on quite a few occasions. Another challenge the café faces is that even though the owner keeps a very good record of daily sales figures and expenses, they don’t do any structured analysis of this data to evaluate costs and prevent avoidable costs.

To address these challenges, I will be applying quantitative analysis techniques such as time-series forecasting to identify sales trends and patterns across different days and weeks. Additionally, I will perform expense analysis to establish relationships between daily costs and sales volumes, providing insight into the café’s cost structure and potential areas for optimization. These analyses will enable me to give recommendations on inventory planning and cost control strategies to minimize wastage and enhance overall profit margins.

2	Organization			Background	
Business	Name:			Coffee	Beans
Address:	DDA	market,	Anand	Vihar,	Delhi-110092
Owner’s Name: Mr. Swastik					

The business that I am working with is Coffee Beans; which is a café based in Anand Vihar, Delhi. Coffee Beans was launched very recently in February of 2025 by Mr. Swastik, who has a lot of experience running his previous café and restaurant ventures. The café sells a variety of products like different types of coffee, garlic breads, snacks, cold drinks, sandwiches and etcetera. The main aim of the café is to provide quality food and beverages to the local community at an affordable price. Swastik’s previous experience in the food industry specifically with cafes has enabled him to create a good experience for his customers. The café had a good start serving around 100 orders a day. Some of the major SKUs are veg momos priced at 80 rs per plate, paneer momos priced at 100 rs per plate, garlic bread priced at 120 rs and cold coffee priced at 100 rs. The café operates from 12:30 pm to 10:30 pm every day.

3 Problem Statement

- 3.1 Due to the unpredictable nature of daily sales, the owner finds it difficult to estimate how much income the café will make on a monthly or even weekly basis. Even though proper records of daily sales and expenses are being maintained, the inconsistency in customer footfall makes it hard to plan ahead. This often leads to either buying too many raw materials which may go unused or run short when demand is unexpectedly high.
- 3.2 Although the café keeps track of all its sales and spending, there is no clear way to figure out where money is being spent unnecessarily. Without analyzing these records properly, the owner is missing out on opportunities to reduce extra costs and improve overall profits.

4 Background of the Problem

The problems faced by Coffee Bites mainly come from the uncertainty that is natural in the food and beverages business. Unlike grocery shops or vegetable vendors where people buy daily as a need, visiting a café is more of a choice or a leisure activity, so the number of customers keeps changing every day. Because of this, the owner finds it difficult to predict sales and plan things like raw material buying or how much to prepare in advance.

Internally, even though the owner keeps proper records of daily sales and expenses, this data hasn't really been used for any useful analysis till now. There is no system to check which days do well, how much is actually being spent unnecessarily, or where costs can be reduced to improve profits. Without looking at these patterns properly, it becomes hard to make decisions that can actually improve the business in the long run.

Externally, factors like weather, holidays, or local events also affect the number of customers visiting the café, but these are not in the control of the owner. All these things make sales uncertain and difficult to predict, which directly affects how much the café should spend on materials or stock each day.

5 Problem Solving Approach

5a. Methods used with their Justifications:

Quantitative

Approaches:

- 1) Time-Series Analysis: I will study the daily sales data over time to figure out which days usually get more customers and which days are slow. This will help in planning raw material purchases better, so that the café doesn't end up buying extra stuff that goes to waste or miss out on sales because of running out of stock.
- 2) Statistical Analysis: I'll use simple stats like average, maximum, minimum, and standard deviation to see how much the sales and expenses change every day. This will give a clear idea of how stable or unstable the café's daily performance is.
- 3) Expense Pattern Analysis: I will look at the daily expense data to see where most of the money is going, like raw materials, delivery, packaging and etc. This will help in spotting areas where costs can be reduced or better managed to save money and improve profits.

Qualitative Approaches:

- 1) Stakeholder Interviews: Engaging in conversations with the café owner and staff will provide insights into operational challenges and customer preferences, complementing our quantitative findings.
- 2) Benchmarking: We'll compare the café's performance metrics with industry standards or similar establishments to identify areas for improvement and best practices.

5b. Data Collection:

- 1) Daily Expense Data: Collect data about where all is the café spending money like raw ingredients, electricity, packaging and staff costs. This will be used to understand the cost of having a delivery boy and extra staff.
- 2) Daily Sales Data: Collect data regarding the total sales in a day and which item sells the most. This will be used to find out which items are sold the most to stock up more raw materials for them. This will also be used to understand profits.

Justification: Focusing on key items sold will help reduce redundant costs that would've been associated with the items that aren't sold as frequently. Understanding profits and expenses will help us understand where expenses can be reduced to increase profits.

5c. Analysis Tools:

- 1) Google Sheets and Excel: For preliminary data processing, cleaning, and analysis. Drawing charts and conducting various computations to uncover meaningful insights.
- 2) Python with Pandas and Matplotlib/Seaborn library: Will be used alongside Excel to explore the data more comprehensively, create visual representations, and extract additional insights.

6 Expected Timeline

6.1 Work Breakdown Structure:

- Data Collection: I collected daily sales and expense data for the café over a period of 60 days in March and April.
- Data Cleaning: I cleaned the collected data by the end of April by removing errors, missing values, and duplicates to make it ready for analysis.
- Proposal – Initial Draft: I prepared and submitted the first version of the proposal in early May.
- Proposal – Revision and Resubmission: I will revise the proposal based on the feedback and resubmit it by mid-June.
- Exploratory Data Analysis: I will start exploring the data in late June by checking overall trends, finding averages, maximums, minimums, and spotting unusually high or low sales days.
- Time-Series & Expense Analysis: I will perform this analysis alongside EDA in late June to find sales patterns, check expense patterns, and compare both to spot areas where costs can be reduced.
- Mid-Term Report Preparation: I will prepare the mid-term report based on the analysis and initial findings by last week of June to first week of July.
- Insights and Recommendations: I will generate insights and prepare simple suggestions to reduce wastage and unnecessary costs along with making basic graphs and visuals by end of July.
- Final Analysis and Result Preparation: I will finish the final round of analysis, polish the results, and finalize all insights by early August.
- Final Report Submission: I will complete and submit the final project report with all the results, visuals, and recommendations in August.

6.2 Gantt chart

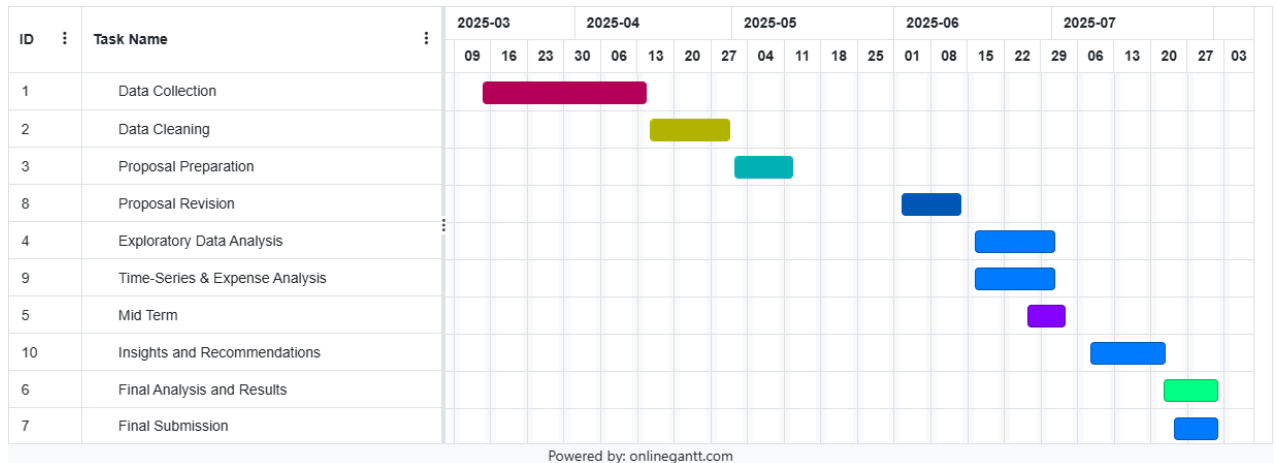


Figure 1 Expected timeline for completion of project.

7 Expected Outcome

- 7.1 To help the café owner understand sales patterns which days usually have higher or lower sales, so that purchases of raw materials can be planned better and wastage can be reduced.
- 7.2 To give the owner a clear view of how daily expenses are behaving compared to sales, and to highlight where unnecessary or avoidable costs are happening.
- 7.3 To provide simple, data-backed suggestions that can help the café control costs and improve profits without risking shortages or overstocking.
- 7.4 To make the café's operations more data-driven, so that future decisions on stock buying and cost management are based on actual sales and expense trends, not guesswork.